Religious Affiliation and Religiosity and Their Impact on Scientific Beliefs in the United States

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Among the greatest hindrances to the acceptance of evolution are religious factors. In this study, we take a more rigorous and detailed approach to the specific effects of religious affiliation and self-reported religiosity on the acceptance of creationist viewpoints and the overall acceptance of evolution. Over 700 religious adults who claimed affiliation with one of four broad religious groups (Southern Baptist, Catholic, Jewish, and The Church of Jesus Christ of Latter-day Saints) were surveyed nationwide using a newly validated instrument. Using structural equation modeling and variable association analysis, we demonstrate that, although creationism and evolution acceptance are negatively correlated, the relationships differ with respect to religious affiliation and with respect to religiosity. These data provide compelling evidence that evolution is viewed differently on the basis of both religious affiliation and religiosity, and they lead to important educational implications when approaching religious students on the subject of evolution.

Keywords: evolution, acceptance, education, religion, religiosity

Despite overwhelming scientific evidence in favor of the theory of evolution, in comparison to 34 other countries, the United States is next to last when it comes to acceptance of the theory (Miller et al. 2006). The US population displays a large degree of variance in its understanding and acceptance of evolution among students and the general population (Miller et al. 2006, PRC 2013). Religious belief has been found to correlate with the low acceptance rate (Miller 2008). However, individuals from different religions accept evolution and its major components to varying degrees on the basis of their respective religious doctrines or cultures. Among Christians, there is a broad spectrum of evolution acceptance, ranging from 58% of Catholics to only 8% of Jehovah's Witnesses (Miller 2008). Therefore, it appears that religious affiliation plays a significant role in evolution acceptance. This study aims to use a systematic and rigorous approach to look more deeply at the nuanced and differential effects of religious affiliation coupled with a person's reported religiosity on their acceptance of creationist viewpoints and rejections of evolutionary theory. To our knowledge, the detail provided in this model is novel and provides a significant contribution to our overall understanding of evolution acceptance among religious students in America.

Religious affiliations differ in evolution acceptance

To explore factors beyond religious affiliation, we examined individuals affiliated with two major religions in America: Judaism and Christianity. Within Christianity, we further focused on three denominations: Catholic, the Church of Jesus Christ of Latter-Day Saints (CJC), and Southern Baptist. We chose these four religious designations because they share a similar creation story but differ in the degree to which they accept evolutionary theory, both doctrinally and culturally. In a national poll, Catholic and Jewish individuals were the most accepting of the theory of evolution (58% and 77%, respectively), whereas the CJC (22%) and Southern Baptists (24%) tended to be the least accepting (Miller 2008). To facilitate the interpretation of the data that will be presented in the present article, we provide below a simple and broadly stated outline of the religious doctrines on evolution for each religion, along with relevant references for further information. However, it is certainly erroneous to assume that all individuals within a given affiliation adhere to—or are even aware of—the established doctrine in their denomination. But the doctrinal position gives us at least an idea of the potential influences involved in a person's acceptance of evolution (see Martin 2010 for a more detailed review).

Judaism. Most research shows that neither Jewish traditions nor Jewish doctrine have major conflicts with the theory of evolution (Sternman 1994, Gitig 2005, Cherry 2011; see Steinberg 2010 for a detailed review). Furthermore, Artson (2011) described Judaism and evolution as "coevolving"

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toward a greater degree of compatibility. There is, however, some marked evolution rejection among the ultra-Orthodox Jewish traditions (see MJL 2018). Interestingly, Judaism and Christianity share a creation story; however, Jewish respondents, on average, accepted evolution at a level that exceeds any Christian denomination by nearly 20% (Miller 2008).

Catholicism. In the 2008 Pew Forum survey, the Catholic respondents were the most accepting of evolution among the Christian denominations we surveyed in this study (Miller 2008). This may potentially be due to statements of neutrality or acceptance made by key leadership in the Catholic church. In 1950, Pope Pius XII made a statement of neutrality toward evolution that left parishioners free to accept the theory if they desired (Mislin 2012). In a 1996 statement, Pope John Paul II stated that evolution has been shown to be "more than a hypothesis" (Martin 2010). More recently, Pope Francis stated that, "evolution… is not opposed to the notion of creation" (McKenna 2014). This would suggest that, at least doctrinally, the Catholic leadership is not opposed to the theory of evolution. Members, however, may hold widely different views (as evidenced by a 42% rejection rate; Miller 2008).

The Church of Jesus Christ of Latter-Day Saints. The CJC church has an impassioned history of religious leaders split on their opinions of the theory of evolution (see Evenson and Jeffery 2005 for a review). This has led to the rejection of the theory by a majority of its members (52% for human evolution as of 2014; PRC 2014). However, the official doctrine on evolution is neutral: "The scriptures tell us why man was created, but they do not tell us how" (Ludlow 1992). The latest explanation appeared in a church publication in October of 2016, stating that "The Church has no official position on the theory of evolution.… Nothing has been revealed concerning evolution" (New Era Magazine 2016, p. 41). The official doctrine affirms a creation story and holds that "God directed the creation of Adam and Eve and placed their spirits in their bodies" (New Era Magazine 2016, p. 41).

Southern Baptists. Statements by R. Albert Mohler Jr, president of the Southern Baptist Theological Seminary, suggest an incompatible viewpoint, such that evolution and Christianity cannot coexist, although certain instances of change via natural selection might be plausible (see Van Biema 2005). Southern Baptist leaders have even gone so far as to support intelligent design as their creationist viewpoint (Southern Baptist Convention 1982). In general, the Southern Baptist leadership has taken a decidedly negative stance against evolution (Weeks 1999, Lemke 2012). However, again, it does not preclude the possibility that many members do not ascribe to the same beliefs as their leadership.

As was described above, there is a disparity of stated religious doctrines toward evolution within religions. In addition, there is great disparity between individuals of any given religion on their knowledge of, understanding of, or ascription to the stated religious doctrines of their religion. This may be an indication that religious affiliation is only one factor that influences the acceptance of evolution. We propose two additional potential factors that may influence the acceptance of evolution: religiosity (as measured by a person's self-reported religious practice, hope, and influence) and acceptance of creationist statements, as we will describe below.

Religiosity predicts evolution acceptance

Religion has been shown to be a significant predictor of the level of acceptance of evolutionary theory (Mazur 2004, Evans 2011, Keeter et al. 2011, Baker 2013). However, the degree to which religion plays a role in one's life may have a differential effect. Therefore, we propose that one influential factor on evolution acceptance is religiosity or religious devotion. This can be and has been defined in many different ways, although it is commonly defined by the degree to which individuals participate in religious practices. For example, both Baker (2013) and Barone and colleagues (2014) measured religiosity as the frequency of church attendance and found that factor to be a negative predictor of evolution acceptance. Baker (2013) and Colburn and Henriques (2006) found a negative correlation between biblical literalism and the acceptance of evolution. Hill (2014) took a more sophisticated view of religiosity, including measures of religious service attendance but also including the frequency of prayer and scripture study, the perceived importance of faith in the respondents' everyday lives, and their feelings of closeness to God. Hill grouped these together as a measure of religious devotion and found it to be a negative predictor of evolution acceptance.

In the case of the present study, we use a method similar to that of Hill (2014) and a modified version of a published inventory of religiosity (Sethi and Seligman 1993). We define *religiosity* as the degree to which someone is involved in organized religious activity (religious practice), the degree to which their religion influences their behavior (religious influence), and the degree to which a person feels hope in a religious sense (religious hope).

Creationist Views influence evolution acceptance

A second proposed influential factor is the acceptance of creationist statements—that is, statements about the interpretation of scriptural accounts of creation. Specifically, much of the controversy about the acceptance of evolution stems from how different religions interpret scripture focused on the creation. Some interpret the scriptural account of the creation literally and believe that the Earth and all its inhabitants were created in a 6-day period—that is, strict creationists. Others interpret the account of the creation more figuratively—for example, suggesting the creation took place in six 24-hour periods with long gaps in between, reasoning that the Earth could have been created in

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six undetermined periods of time over millions to billions of years, or positing that the *Genesis* account is entirely figurative and was not meant to be taken as a historical account of creation. Individuals who are strongly committed to a more figurative interpretation are more willing to accept the theory of evolution than are those who take the scriptural account literally (Levesque and Guillaume 2010, Lloyd 2014), and strict creationists tend to be adamantly opposed to evolution (Scott 1997).

Creationist beliefs and evolution are not necessarily mutually exclusive. For example, one might firmly believe that God had a hand in creating life but still firmly accept that evolution was the mechanism through which this creation occurred. We suggest that religious affiliation and religiosity may affect the acceptance of creationist statements, and the acceptance of these statements may, in turn, influence the acceptance of evolution as a valid theory.

Components of evolution acceptance

In this study, we sought to determine the interactive effects of these three factors (religious affiliation, religiosity, and acceptance of creationist statements) on the acceptance of evolution. We sought to determine the relationship between these three factors and the acceptance of four components of evolutionary theory: general definitions of evolution (i.e., gradual change in species over time), the age of the Earth, the process of natural selection, and the origin of human beings. Given the variations in how individuals interpret the creation story, one might expect to see variation in the acceptance of these components of evolution. Many students accept microevolution (i.e., gradual change in species over time), leading to an acceptance of general definitions of evolution and of the process of natural selection, but oppose macro-level evolution that would include the speciation of hominids (Mead et al. 2017). In addition, the accepted age of the Earth is likely to vary with individuals on the basis of their interpretation of the length of the "time periods" of creation, leading those who take a more figurative approach to be more likely to accept an old Earth, whereas those taking a literal approach may struggle with this concept.

Research hypotheses

The purpose of the present research was to test to what extent religious affiliation and religiosity influence the acceptance of creationist statements and predict the acceptance of evolution. This research adds to our understanding of this issue, as we use a refined and validated scale of religiosity and a more inclusive analysis to explore how multiple factors may work together to influence acceptance of evolution. Specifically, we aimed to test two main hypothesized causal factors—religious affiliation and religiosity (as measured by self-reported religious practice, influence, and hope)—on the acceptance of creationist views and the acceptance of evolution. In addition, we propose a model for the relationship between these variables.

Surveyed Population

To study the relationships between religious affiliation and religiosity and the acceptance of evolution, a Qualtrics Panel was used to survey 724 individuals from the general, nationwide US population. The individuals were chosen on the basis of their stated affiliation with one of four denominations: Catholic (n = 178), Jewish (n = 178), CJC (n = 182), and Southern Baptist (n = 182). We recognize that "Judaism" represents at least three major branches; therefore, our Jewish respondents are likely representative of an average across Jewish traditions. Fifty-seven percent of the individuals surveyed were between the ages of 18 and 25 (i.e., college age). The remaining respondents were older than 25.

Survey instrument

Prior to conducting our survey, we validated the instruments for measuring religious affiliation, religiosity, the acceptance of creationist ideas, and the acceptance of evolutionary theory. The administered instrument was created and validated for the specific purpose of this study. It was intended to measure the participants' religious affiliation, religiosity, creationist views, and evolution acceptance. The instrument was divided into four parts (see the supplemental material):

The first part was demographic questions. The demographic items of religious affiliation, age, and college experience were used as grouping variables in the analyses.

Next, we measured religiosity. One limitation of most previous research focused on evolution acceptance is that religiosity has been measured with a single dichotomous item (e.g., religious or a-religious) or with a nonvalidated scale to determine comparison groups (e.g., Lawson and Worsnop 1992, Village and Baker 2013, Rissler et al. 2014). The best practice is to use a scale instead of a single item to measure a latent variable such as religiosity (Gardner et al. 1998). Multiple-item scales of religiosity have been published (Roof and Perkins 1975, Altemeyer 1988, Sethi and Seligman 1993, Pfeifer and Waelty 1995) but have not been used to measure how religiosity might affect the acceptance of evolution. Research in which a validated religiosity scale was used when testing the relationship between religiosity and the acceptance of evolution is scarce (see Hill 2014 and Manwaring et al. 2015 for exceptions). For this study, we used a modified version of Sethi and Seligman's (1993) religiosity instrument, which has been previously validated (Manwaring et al. 2015). These 15 items (on a six-point Likert scale) were used to measure three indicators that would suggest a person's level of commitment to their religion: personal religious practices, the influence religion has had on decision-making, and how religion influences their outlook on the present and the future (hope).

The responses to the 15 items were summed by subset to calculate religious practices, influence, and hope and then together to compute a measure of each participant's overall religiosity. The respondents were considered to have "high religiosity" if they were greater than one standard deviation

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Table 1. Sample size by religiosity.

	Catholic	Jewish	CJC	Southern Baptist
High religiosity (more than 1 SD above mean)	30	31	23	33
Average religiosity (within 1 SD of mean)	122	126	129	123
Low religiosity (more than 1 SD below mean)	26	21	30	26
Total	178	178	182	182

above the mean, "average religiosity" if they were within one standard deviation of the mean, and "low religiosity" if they were greater than one standard deviation below the mean. Table 1 shows the sample's distribution of religiosity.

We pilot tested this survey among 585 individuals recruited through social media. We preformed construct validity and reliability tests to ensure the quality of the instrument, described below. The scale had a high level of internal consistency, as determined by a Cronbach's alpha of .914. We then measured creationist views. These 13 items measured an individuals' agreement with aspects of biblical creationism. These creationist views included the nature of God, the age of the Earth, the fixity of species, and the origin of humans.

These items and the evolutionist statements described below are not necessarily mutually exclusive. For example, one could answer in the affirmative to the statement, "God created the Earth and all living organisms" without rejecting evolution as a mechanism. Items had six response categories on a Likert scale. The items were focused on how strongly the individuals agreed or disagreed with statements on a general belief in God, the length of creation, the nature or origin of nonhuman species, and the nature or origin of humans. The responses were averaged, with higher scores representing stronger creationist beliefs. We piloted this survey among 585 individuals recruited through social media and 73 individuals recruited in introductory biology classes at the authors' institution. Validity and reliability tests ensured the quality of the instrument as measured by confirmatory factor analyses and Cronbach's alpha (coefficient = .95; details described below).

Finally, we measured evolution acceptance. These items were used as a measure of the dependent variable—acceptance of evolution. The components of evolution addressed included general definitions of evolution (i.e., gradual change in species over time), the age of the Earth, natural selection, and the origin of humans. These 12 items were answered on a six-point Likert scale. The responses were averaged, with higher scores representing stronger acceptance of evolution. This survey was piloted using the same 585 individuals recruited through social media as used for the measure of creationist views. Validity and reliability tests demonstrated the instrument measured a single factor (via confirmatory factor analysis, described below) and had a high level of internal consistency with a Cronbach's alpha score of .953.

Statistical approach used

The survey responses were analyzed in relation to each other using an analysis of variance (ANOVA) in SPSS version 21 (IBM, Armonk, New York).

Instrument validation. To investigate the validity and reliability of our instruments, we used exploratory factor analysis, confirmatory factor analysis, and structural equation modeling. An exploratory factor analysis was performed on each instrument in SPSS version 21. A scree plot, eigenvalues, and percentage variance explained were examined to determine how many factors to retain. The confirmatory factor analysis and model fit analyses were then performed on each instrument in Mplus in order to validate the instruments and confirm goodness of fit to the data. Items were removed or adjusted according to results in order to provide the strongest measures of each latent variable. The fit of each instrument model to the data was analyzed via fit indices, TLI, CFI, RMSEA, χ^2, and *p*-value statistics. In the confirmatory factor analysis and measurement model for each instrument, the items were classified as categorical to activate the weighted least squares estimation instead of the default maximum likelihood estimator. All of the instruments were then included in a full measurement model to determine the model that best fit the data, given the factors being addressed. We tested for scalar and factor invariance between religions in the measurement model, using χ^2 difference testing between nested models.

Construct validation. An exploratory factor analysis rendered three factors from our religiosity measure with eigenvalues above 1 that also explained over 5% of the variance seen in the data. The resulting religiosity factors were designated *practice*, *influence*, and *hope*. One factor resulted for each of the remaining measures of creationist views and evolution acceptance. Each of these measures had only one factor with an eigenvalue higher than 1 that also explained more than 5% of the variance. We then used confirmatory factor analyses to determine the validity of each construct as measured by individual instruments. The factor loadings for each item from all instruments were high (above .5), with few exceptions. The fit statistics for each confirmatory factor analysis can be found in table 2. Our religiosity scale again factored into three variables: practice, influence, and hope. Items were removed from the religiosity and creationist views instruments (e.g., Q2.4: "How often do you pray?" Q3.2: "All creatures on Earth were created in the last 10,000

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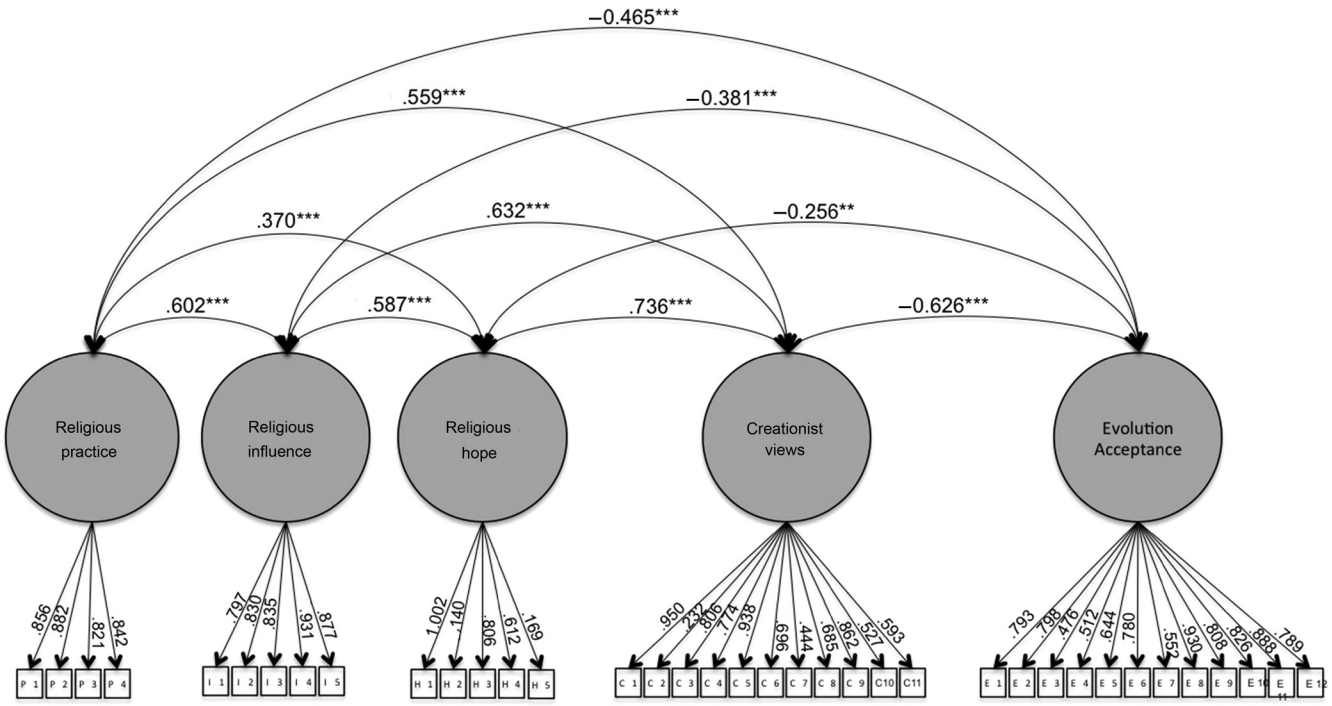


Figure 1. Correlations between factors and the loadings of each item on their respective factor. Unidirectional arrows show factor loadings for each item and correlational relationships are show by bidirectional arrows; both are labeled with standardized coefficients. Significant relationships are marked: *p < .05. **p < .01. ***p < .001.

years," and Q3.2: "All present day humans are direct descendants of Adam and Eve.") because of redundancy (i.e., a lack of uniqueness) or poor fit. The fit statistics show that each instrument model fit the data well.

Measurement modeling. The complete measurement model can be seen in figure 1. The measurement model shows correlations between factors and the loadings of each item on their respective factor. When we tested for invariance in the model, we found scalar invariance and partial metric invariance. An adjusted metric model was accepted. The only adjustment to the model was not allowing the thresholds of one religious practice item (P4) to be freely estimated; attempting to freely estimate the thresholds of P4 caused the analysis to terminate early because of convergence problems. The overall fit of the measurement model was good (see table 2).

Religious affiliation affects acceptance

To test the effect of religious affiliation on the acceptance of creationist statements and evolutionary theory, we compared the survey responses among individuals of different religious affiliations. A total acceptance score for each construct was summed for each survey respondent. A one-way ANOVA showed that the acceptance of creationist statements was significantly different among the religious affiliations ($p < .0001$; see table 3 and figure 2a), being highest among the Southern Baptists and CJC, who were not statistically different from one another, and lowest among the Catholic and Jewish individuals. An opposite pattern was shown for the acceptance of evolution among religion affiliations ($p < .001$; see table 4 and figure 2b). Baptists and CJC were the least accepting of evolution—once again, not statistically different from one another—whereas Catholic and Jewish individuals were the most accepting.

The respondents were then split down the middle on the basis of their responses, such that the measure of acceptance of creationist statements became dichotomous (i.e., those in the upper half of the data were considered "accepting" and those in the lower half of the data were considered "rejecting"). Counts of how many accepted and rejected creationist statements within each religious affiliation are shown in table 3. It is clear that the Southern Baptist and CJC populations in our study were more accepting of creationist statements, whereas the Catholic and Jewish populations were generally more rejecting of creationist statements. From a similar grouping for acceptance of evolution, our data show that the Catholic and Jewish populations were generally more accepting of evolution, whereas the Baptists and CJC were generally more rejecting of evolution (see table 4).

To determine whether general acceptance differed significantly among various components of evolutionary theory (general evolution, time, natural selection, and human evolution), we ran independent-samples *t*-tests comparing the acceptance levels. To test whether religious affiliation

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Table 2. Fits statistics for individual factors and measurement model.

				Chi-square test			
	TLI	CFI	RMSEA	χ2	Degrees of freedom	p-value	Adjustments
Religiosity	.977	.970	.081	1057.23	481	< .001	One item (Q6) removed. Three factors.
Creationist views	.992	.981	.103	249.56	86	< .001	Two items (Q10 and Q12) removed
Evolution acceptance	.989	.957	.122	361.919	98	< .001	None
Complete measurement Model	.972	.956	.111	953.410	295	< .001	Thresholds were estimated freely except for P4.

Table 3. Percentage of individuals rejecting or accepting strict creationist views for each religious affiliation.

	Overall rejecting		Overall accepting	
Religion	Count	Percentage	Count	Percentage
Baptist	38	20.9	144	79.1
Catholic	103	56.6	79	43.4
Jewish	135	75.8	43	24.2
CJC	54	29.7	128	70.3
Total	330	45.6	394	54.4

Abbreviation: CJC, Church of Jesus Christ of Latter-Day Saints.

differentially affected the acceptance of various components of evolutionary theory, one-way ANOVAs were conducted comparing the acceptance of each component within each religious affiliation. The data were approximately normally distributed for each group, as assessed by visual inspection of Normal Q–Q plots**.** Regardless of religious affiliation, the individuals were significantly more accepting of an old Earth than any other component. However, the religious affiliations differed widely in their acceptance of human evolution (see figure 3).

Within each component of evolution, there was a significant main effect of religious affiliation on acceptance (see table 5). The Catholic and Jewish individuals had higher acceptance of all four components of evolution than did the Baptist and CJC individuals. Games-Howell post hoc analyses revealed that most comparisons among religious affiliations on the acceptance of specific components were significant (see table 5).

Religiosity affects acceptance

To test the effect of religiosity on the acceptance of creationist statements and evolutionary theory, we compared the survey responses among individuals of different religiosity. Among all the religious affiliations investigated in the present article, an ANOVA showed that the acceptance of evolution significantly differed with religiosity ($F(1,721) = 30.221$, $p < .001$). The acceptance of statements about evolution was highest among the individuals of low religiosity (mean (M) = 4.60, standard deviation (SD) = 1.35), followed by individuals of average religiosity (M = 4.09, SD = 1.16), then individuals of high religiosity (M = 3.39, SD = 1.35). A Games-Howell post hoc analysis revealed that all of the comparisons among all levels of religiosity on the acceptance of evolution were significant ($p < .001$).

We tested for an interaction between religiosity and religious affiliation on the acceptance of evolution or creationist views (i.e., whether religiosity functions differently in influencing the acceptance of evolution or creationist statements depending on religious affiliation). A one-way ANOVA revealed a significant interaction between religious affiliation and religiosity on the acceptance of evolution ($F(6,724) = 6.037$, $p < .001$), meaning that religiosity affected evolution acceptance differently depending on the respondents' religious affiliation. Among the Jewish individuals, those of average religiosity accepted evolution as much as those having low religiosity, whereas, among the other religious affiliations, those of average religiosity were more likely to group with highly religious people in their level of acceptance of evolution (see figure 4a). There was no significant interaction between religious affiliation and religiosity on the acceptance of creationist statements (see figure 4b).

A proposed model of relationships

Using structural equation modeling, we built and tested a model of the relationships among the three religiosity subsets (practice, influence, and hope), the acceptance of creationist statements, and the acceptance of evolution, allowing for variation among religious affiliations. Structural equation modeling is a robust statistical analysis method used to examine the relationships among multiple variables simultaneously (Muthen and Muthen 2012). We used Mplus version 7.3 to analyze the relationships of the variables for this study. The variables analyzed in our model were religiosity (measured by religious practices, religious influence,

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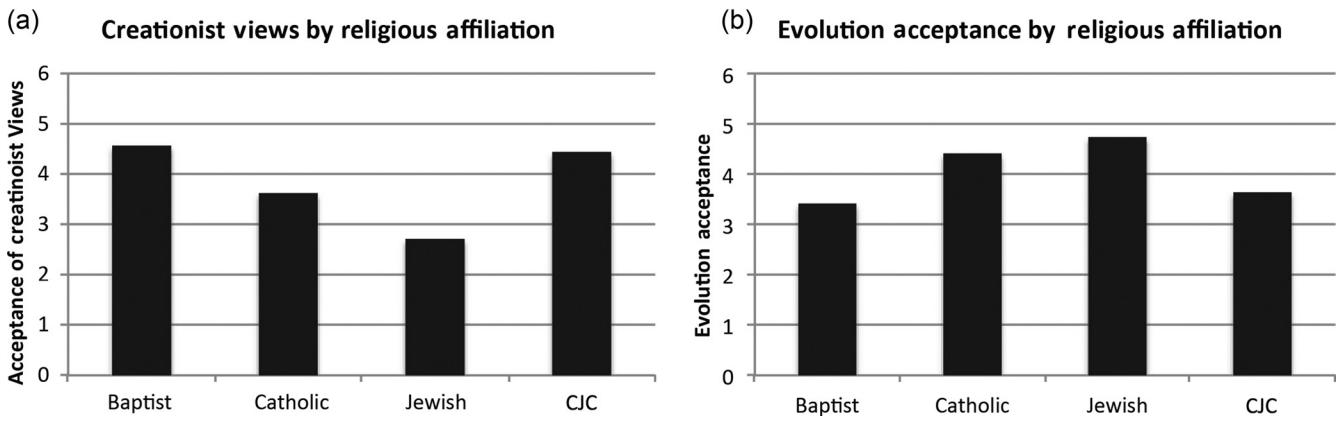


Figure 2. Comparison of acceptance of creationist views and evolution acceptance between religious affiliations, on a six-point Likert scale. (a) The average acceptance of creationist views by religious affiliation. (b) The average acceptance of evolution by religious affiliation. Abbreviation: CJC, Church of Jesus Christ of Latter-Day Saints.

and religious hope), the acceptance of creationist statements, and the acceptance of evolution. The variables were grouped by religious affiliation. We developed and tested a hypothesized path diagram (structural model) for these variables. The structural model was run in Mplus, and we compared how it functioned among groups (religious affiliations). We tested for three kinds of invariance (structural weights, structural covariance, and structural residuals) in our model using χ^2 difference testing between nested models.

To analyze the relationships between latent variables, we tested our hypothesized model (see figure 2). The model is a good fit for the data as was indicated by fit statistics and *p*-values (TLI = 0.941, CFI = 0.976, RMSEA = 0.104, χ^2 = 825.001, $p < .001$). The vast majority of parameter estimates were statistically significant. When we tested for structural invariance, the best-fitting model allowed religious hope to differ for the Jewish individuals and religious practice to differ for the Southern Baptist individuals. In other words, religious hope was a larger predictor of creationist views for Jewish individuals than for other affiliations, and religious practice was only a significant predictor of creationist views for Southern Baptists. The final model shows that religiosity was a significant predictor of the acceptance of creationist statements and that religiosity and the acceptance of creationist statements were significant predictors of the acceptance of evolution. In other words, the higher a person's religiosity was, the more likely they were to agree with creationist statements, and the more likely a person was to agree with creationist statements, the less likely they were to accept evolution (figure 5).

Interpretations and practical applications of the effects of religious affiliation on acceptance

Our results on the acceptance of evolutionary theory align with those found in the 2008 Pew Survey (Miller 2008). In the Pew Survey, 58% of Catholics, 77% of Jewish people, 22% of CJC, and 24% of Southern Baptists were found to accept evolution; we report values of 70%, 81%, 35%, and 31%, respectively. This may be because more than half of our study population was college age. Recent research suggests that religiosity among 18–29-year-olds has dramatically decreased (see Pomeroy 2016). Given the robust relationship between religiosity and a rejection of evolutionary theory, this may also affect the increase in acceptance rates.

Interestingly, the acceptance rates did not consistently mirror the official doctrines on evolution of each religion. Our study showed that the acceptance of creationist statements was in direct opposition to the acceptance of evolution for all of the religions studied. This is not surprising, given that many of the creationist statements seem to be in direct opposition to evolutionary viewpoints. However, some statements, such as holding a belief in creation being directed by God, do not exclude an acceptance of evolution. Individual creationist statements, however, were not analyzed in the present model. Furthermore, research into the relationship (or lack thereof) between a belief in God as the creator and the acceptance of evolution among religious individuals would be fruitful. Not surprisingly, the Jewish and Catholic individuals were the least likely to accept creationist statements and the most likely to accept evolution.

Research suggests that the Jewish religion has not been opposed to evolutionary theory (Sternman 1994, Gitig 2005, Steinberg 2010, Cherry 2011). Pope Pius XII was the first Catholic Pope to openly consider evolution as a means for creation (1950), and subsequent Popes have made similar statements (Mislin 2012, McKenna 2014). Although 42% of Catholics still reject evolution (Pew Research Center 2008), most religious individuals in the United States take on a more figurative interpretation of the creation of the Earth (Jones 2011). In fact, previous research shows that many Judeo-Christian clergymen find that science and religion are compatible, if not complimentary (Colburn and Henriques 2006, Zimmerman 2010). The CJC has a more figurative interpretation of the

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Table 4. Percentage of individuals rejecting or accepting evolution for each religious affiliation.

	Overall rejecting		Overall accepting	
Religion	**Count**	**Percentage**	**Count**	**Percentage**
Baptist	125	68.7	57	31.3
Catholic	55	30.2	127	69.8
Jewish	34	19.1	144	80.9
CJC	119	65.4	63	34.6
Total	333	46	391	54

Abbreviation: CJC, Church of Jesus Christ of Latter-Day Saints.

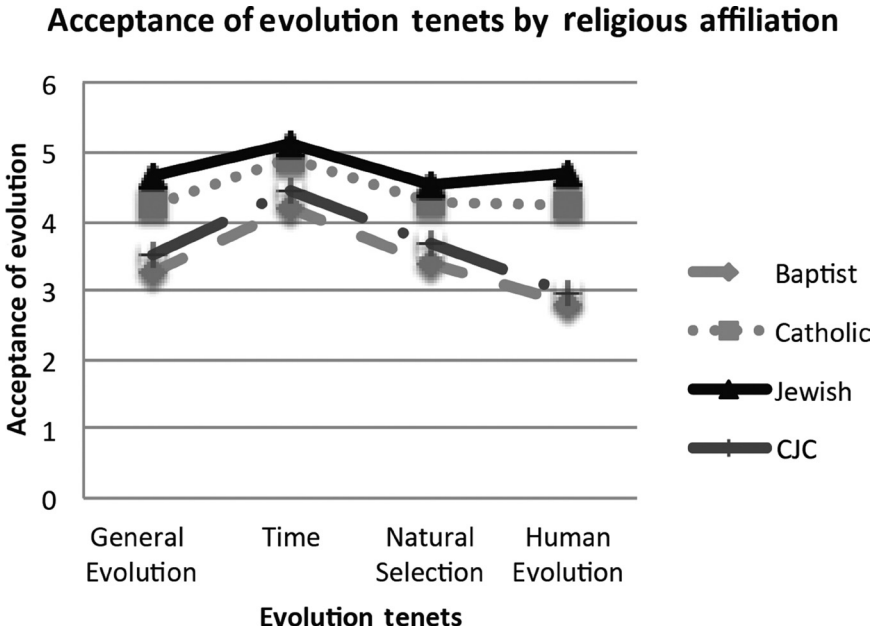


Figure 3. Acceptance of the tenets of the theory of evolution by religious affiliation. The acceptance of evolution instrument was divided into four components. The average acceptance of each component is depicted within each religious affiliation. Abbreviation: CJC, Church of Jesus Christ of Latter-Day Saints.

Bible and has taken no stance on creationism being taught in the classroom. Their doctrine states that God created the Earth, but there is no clear doctrine as to the mechanisms behind the creation (Ludlow 1992), leaving the church in a relatively neutral position toward the theory of evolution. However, the majority of our CJC respondents rejected evolutionary theory.

Not surprisingly, the Baptists were the most likely to accept creationist statements and the least likely to accept evolution. Many Baptist church leaders support a literal interpretation of the creation account in the Bible (Scott 1997). In 1982, the Southern Baptist Convention declared, "Whereas, creation-science can be presented solely in terms of scientific evidence without any religious doctrine or concepts…[we] express our support for the teaching of Scientific Creationism in our public schools" (Southern Baptist Convention 1982).

Within creationist movements, one of the largest battles regarding evolution has concerned the age of the Earth (Reed 1998, Reed and Williams 2011, Doyle 2012). Young Earth creationism is based on the belief that the Bible is meant to be a historical text with specific details about a 6-day creation. (Walker 2000, Humphreys 2012). Surprisingly, our data show this component of evolution to have been the least controversial among all of the religious affiliations sampled. This may reflect a turning point among literal creationists and may be due to the changes in belief systems of the rising generation (see Pomeroy 2016). With the rise of the information age comes the easy and almost universal access to scientific information. This may be a driving force behind the increased acceptance of geological dating methods and the evidence that supports an Earth that is at least 4.6 billion years old.

Human evolution, on the other hand, remains one of the largest controversies hindering the acceptance of evolutionary theory (PRC 2013). Our data support this trend. In addition, our data suggest that this component may be the driving force of the variation among religious affiliations in acceptance rates. The dichotomy seen between Catholic and Jewish individuals, who are much more accepting, and CJC and Southern Baptist individuals, who are much less accepting, can be seen distinctly in this one controversial issue. It suggests that varying doctrines on human origins within each affiliation may be the major determinant of acceptance of evolution.

Interpretations and practical applications of the effects of religiosity on acceptance

Our results show that, in terms of self-reported religiosity (i.e., religious practice, influence, and hope), members of the CJC ranked highest, followed by Southern Baptists, then Catholics, then Jewish people. This corresponds to

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Table 5. Acceptance of components of evolution by religious affiliation.

		Baptist (B)		Catholic (C)		Jewish (J)		CJC		
Component	F(3,720)	Mean (*M*)	95% confidence interval (CI)	*M*	95% CI	*M*	95% CI	*M*	95% CI	Games-Howell post hoc analyses
General evolution	54.70***	3.25	0.19	4.23	0.17	4.67	0.17	3.50	0.15	B–C: $p < .001$
										B–J: $p < .001$
										B–CJC: NS
										C–J: $p < .001$
										C–CJC: $p < .001$
										J–CJC: $p < .001$
Time	24.45***	4.21	0.19	4.90	0.14	5.11	0.16	4.45	0.15	B–C: $p < .001$
										B–J: $p < .001$
										B–CJC: NS
										C–J: $p < .001$
										C–CJC: $p < .001$
										J–CJC: $p < .001$
Natural selection	35.39 ***	3.39	0.19	4.29	0.17	4.51	0.18	3.67	0.16	B–C: $p < .001$
										B–J: $p < .001$
										B–CJC: NS
										C–J: NS
										C–CJC: $p < .001$
										J–CJC: $p < .001$
Human evolution	79.19***	2.82	0.23	4.23	0.20	4.68	0.20	2.95	0.20	B–C: $p < .01$
										B–J: $p < .01$
										B–CJC NS
										C–J: $p < .01$
										C–CJC: $p < .01$
										J–CJC: $p < .01$

Abbreviation: CJC, Church of Jesus Christ of Latter-Day Saints. ***p < .001.

the pattern of acceptance of evolution among religions. Not surprisingly, the levels of religiosity inversely corresponded to levels of acceptance of evolutionary theory; that is, the higher a person's religiosity was, the more likely they were to reject evolutionary theory, regardless of their affiliation. However, this relationship was not consistent within religious affiliations. Among those of Jewish affiliation, the levels of acceptance among those of low and medium religiosity are equally high. It is not until one claims high religiosity that acceptance levels drop. Whereas, among CJC and Catholic individuals, the trend was the opposite, with those of low religiosity being more accepting than individuals who claim medium or high religiosity. Among Baptists, there was a linear correlation between religiosity and acceptance. Several explanations may account for this observed trend. It is likely that the higher your religiosity, the more exposed you are to the cultural aspects of your religion (i.e., through higher religious practice). This would explain the trend seen among the CJC participants, in which the church doctrine is neutral but church culture takes a negative stance. It is also possible that the higher your religiosity, the more knowledge you have of your own religion's espoused beliefs. However, this would appear to favor the relationship seen among the Catholics and Jewish individuals, and even among the Southern Baptists. But it does not explain the discrepancy seen within the CJC population. It also does not explain the still substantial rejection of evolution among some Catholic individuals. Previous research has suggested that knowledge of doctrine within the CJC on evolution is relatively low, even among highly religious college students (see Manwaring et al. 2015).

Interpreting our model

The model we proposed was that religiosity directly influences the acceptance of creationist statements and that this, in turn, directly influences evolution acceptance. Our model showed a robust fit with the data, showing both correlations and predictive relationships between variables using structural equation modeling. All three religiosity measures (religious practice, religious influence, and religious hope)

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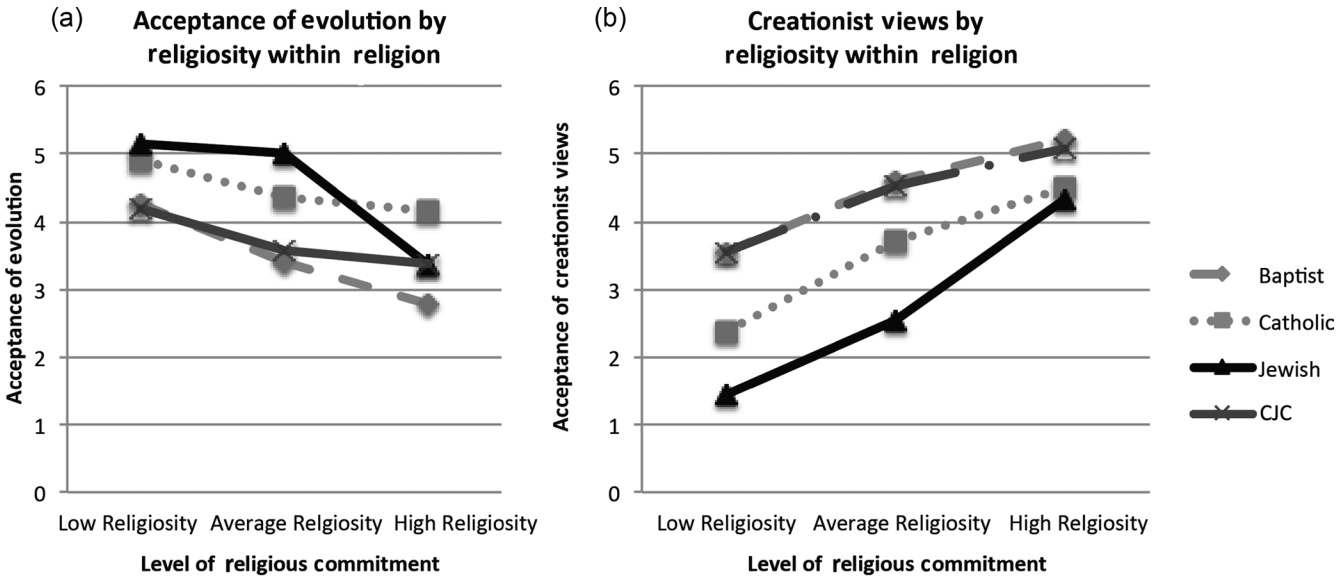


Figure 4. Acceptance of creationist views and evolution by religiosity within each religious affiliation. Religiosity was divided into three bins: "high religiosity" were those who scored greater than one standard deviation above the mean on the religiosity instrument, "average religiosity" were those who were within one standard deviation of the mean, and "low religiosity" were those who scored greater than one standard deviation below the mean on the religiosity instrument. (a) shows acceptance of evolution; (b) shows acceptance of creationist views. Abbreviation: CJC, Church of Jesus Christ of Latter-Day Saints.

correlated with one another, suggesting that each influences the others. This is not surprising, given that the three measures were intended to measure overall commitment to one's religion.

Looking first at religious practice (i.e., church attendance, reading of scripture, time devoted to religious activities), we see that, in the present study, it did not predict an individual's acceptance of creationist statements for Catholic, Jewish, and CJC individuals; however, it did predict the acceptance of creationist statements for Southern Baptists. This may be an indication of the open and frequent teaching of strict creationism in the Southern Baptist religion (Scott 1997). Therefore, individuals who have more frequent church attendance may be more exposed to this doctrine and therefore more accepting of the creationist views taught. Religious practice did, however, negatively influence the individual's acceptance of evolutionary theory. This may be due to the ideals that exist implicitly in religious culture, even if anti-evolution sentiments are not explicitly taught, as in the case of the CJC. In addition, frequent church attendance or practice may lead to an individual taking greater offense when scientific claims seem to imply that God does not exist, thus increasing the chance that they will reject evolution, regardless of the official doctrine of their religion.

Regarding religious influence (i.e., how much influence religion has in decision-making, goals, lifestyle, etc.), the model showed a strong positive correlation with acceptance of creationist statements across all of the religious groups surveyed. It did not, however, have a direct influence on acceptance of evolution but did have an indirect negative influence through creationist views.

Religious hope (i.e., how religion influences one's outlook on the present and the future) was the strongest predictor of the acceptance of creationist statements among Catholics, CJC, and Southern Baptists, and it was even significantly stronger for Jewish individuals. However, when looking at the direct effect of religious hope on evolution acceptance, it appears that more hope correlated with more acceptance of evolution (see figure 5). This relationship is potentially understandable, given that many of the statements in this measure were not necessarily indicative of religiousness (e.g., "Do you believe it is possible for all humans to live in harmony together?"). It may also be that religious hope is least indicative of actual religious culture and is shared by more and less religious and a-religious individuals alike. Certainly, more research on this relationship is warranted.

Conclusions

On the basis of these data, it appears that the acceptance of evolution among religious individuals is a complex issue involving more than just the ascription to a particular religious affiliation. Personal beliefs and convictions often trump official doctrines of the church, whether it is an informed decision or it is due to a simple lack of awareness of these official doctrines. In addition, the level of commitment someone has to their religion may be an important factor in the acceptance of evolution. This is perhaps due to

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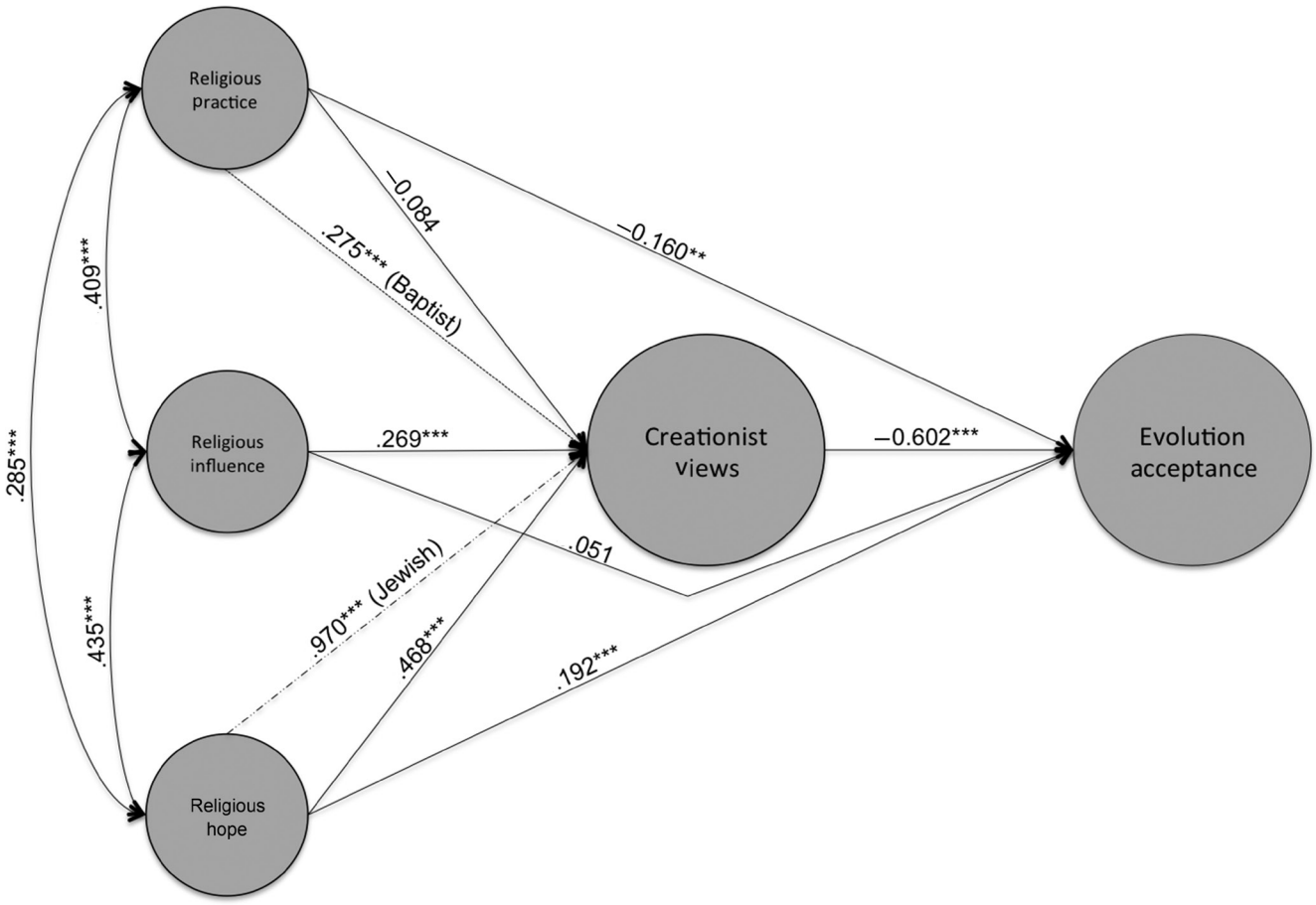


Figure 5. Religious practice, influence, and hope and their contributions to acceptance of creationist views and evolution. Unidirectional arrows show predictive relationships and correlational relationships are show by bidirectional arrows; both are labeled with standardized coefficients. The structural model is the same for all religions, with exceptions displayed with lighter arrows. Significant relationships are marked: *p < .05. **p < .01. ***p < .001.

the increased exposure to religious culture a person would have with increased participation in the religion, a culture that may support the acceptance of creationist statements. Unfortunately, the culture may not match official doctrines. McNamara (1985) points out that often studies on the influence of religion on beliefs and behavior fail to consider *verstehen*, or the ability to interpret meaning of behavior from the actor's point of view. Their intended meaning may differ from the observer dramatically given that people bring their own understanding and meaning with them into decisions they make. Therefore, people may interpret the doctrine of their church in very different ways from each other depending on their own background and belief system. This will necessarily lead to discrepancies that we see between stated ideologies and actual practices by religious constituents (Denton 2004). For example, in a study on contraceptive use in Canada, researchers showed that religious culture is just as important as, if not more important than actual religious doctrine on a women's decision to use contraception (Srikanthan and Reid 2008). Atkin and colleagues (2008) found that when it came to making decisions regarding prenatal screening for potentially lethal genetic disease and decisions to terminate pregnancy, many of the religious respondents suggested that it would be helpful to know their religion's doctrine on the subject but that, ultimately, the decision would come down to their culturally influenced morality. Indeed, Denton (2004) encourages researchers to make a distinction between the official declarations of a religious group, the beliefs of its members, and their actual actions.

Regarding evolution, encouraging a better understanding of one's religion and a greater awareness of official religious doctrine may lead to higher acceptance. In a recent study on college students at a highly religious institution affiliated with the CJC church, researchers showed that addressing doctrinal beliefs, including human origins, directly led to increased acceptance of evolution (Manwaring et al. 2015). An approach similar to this, where students are encouraged to learn official doctrines and consider the controversy in a non-threatening environment may help in overcoming this potential barrier, barring any official doctrinal statements

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prohibiting belief in human evolution, for example, as in the case with Southern Baptists.

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Supplemental material

Supplemental data are available at *BIOSCI* online.

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